

National University of Computer and Emerging Sciences



Lab Manual 12

Programming Fundamentals

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Activities:

Note: Include input validation in programs wherever applicable. Don't use any build-in function.

1. Create a program that manages an employee payroll system using files. The program should:
 - Write employee details to a file:
 - Employee ID, Name, Department, Monthly Salary.
 - Read the file to display all employee records.
 - Allow appending new employee records.
 - Provide an option to calculate and display the total annual payroll for all employees.

Functionality Requirements:

- The file should store data in a structured format.
- Use functions to handle:
 - Writing employee details to the file.
 - Reading and displaying all records from the file.
 - Appending new employee records to the file.
 - Calculating and displaying the total annual payroll.

Sample Input (from user):

Enter Employee ID: 101

Enter Employee Name: Alice

Enter Department: HR

Enter Monthly Salary: 50000

Text File:

ID Name Department Salary

101 Alice HR 50000

102 Bob IT 60000

Sample Output (on console window):

Employee Records:

101, Alice, HR, 50000

102, Bob, IT, 60000

Total Annual Payroll: 1320000

2. Write a program that manages student academic records using file handling. The program should:
 - Create a file and write the following details for multiple students:
 - Roll Number, Name, Subject, and Marks.
 - Read and display all student records from the file.
 - Allow appending new student records to the file.
 - Provide a function to calculate the average marks for a specific subject.

Functionality Requirements:

- Store student data in a formatted file.
- Functions should handle:
 - Writing student details to the file.
 - Reading and displaying all records.
 - Appending new student data.
 - Calculating the average marks for a subject.

Sample Input:

Enter Roll Number: 1

Enter Student Name: John

Enter Subject: Math

Enter Marks: 85

Text File:

RollNo Name Subject Marks

1 John Math 85

2 Jane Math 90

Sample Output:

Student Records:

1, John, Math, 85

2, Jane, Math, 90

Average Marks in Math: 87.5

3. Write a C++ program that reads in data from a file “input.txt” according to the given format and prints the information on screen as shown in the example below.

Format:

The file begins with an integer. This integer tells us the number of students whose record is stored in the file. Following this integer we have the first, middle and the last name of the first student. Then there is an integer which tells us how many assignments did the first student submit followed by the marks. Finally, the list of marks (for the first student) is followed by the student's roll number. Then the file has the first, middle and last name of the second student followed by the number of assignments he/she submitted and so on.

A sample file is given below (there is a space after every number and name).

Note: This is just a sample; your program must work on any file that follows the above format.

3 Ali Ahmad Khan 3 2.4 3.3 7.1 L12-2000 Kiran Ali Panday 2 7.0 6 L12-3000 Imran Javed Khan 1 2.2 L12-3001

For the file given above the program must print the following on screen (Roll No, last name and total marks):

L12-2000 Khan 12.8

L12-3000 Panday 13.0

L12-3001 Khan 2.2